outputs, is vividly illustrated in the first- and last-name distribution bar charts (Figures 2, 3).

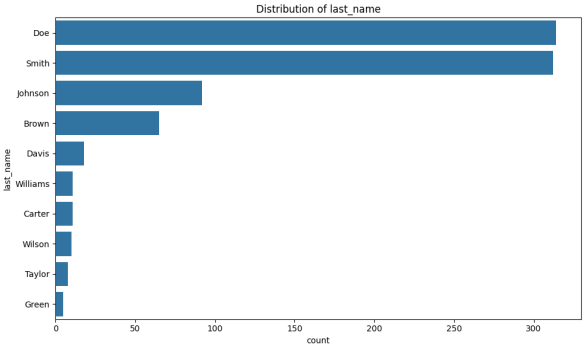

Figure 3: Distribution of Last Names

The exponential frequency decay from these archetypes (e.g., "John" at n ≈300 vs. "James" at n ≈50) reflects LLMs' propensity to default to culturally dominant name tropes. Notably, the long-tail distributions—74.3% of unique first names occurring at <1% frequency—are masked in these charts by the logarithmic scaling of rare categories, emphasizing the tension between statistical representativeness and visual clarity.

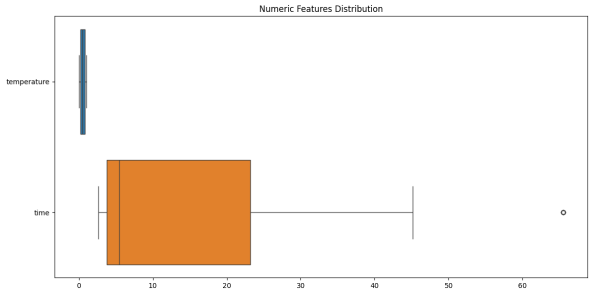

Figure 4: Numeric Distributions

While temperature settings spanned 0–1 uniformly ($\Delta = 0.1$), the narrow IQR of the temperature distribution (Figure 4) reveals clustering around the mean ($\mu = 0.49$), suggesting limited experimental exploration of extreme stochasticity regimes. Paradoxically, the concentration of rare names at intermediate temperatures (0.3–0.7) implies that maximal entropy – and thus creativity – occurs within this constrained range.

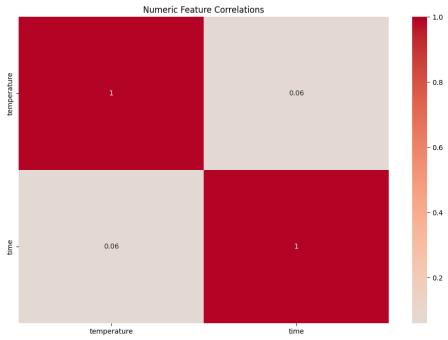

Figure 5: Numeric Correlation

The absence of correlation between temperature and time (Figure 5) further dissociates computational cost from stochastic exploration, challenging assumptions that "hotter" sampling inherently demands more processing resources.

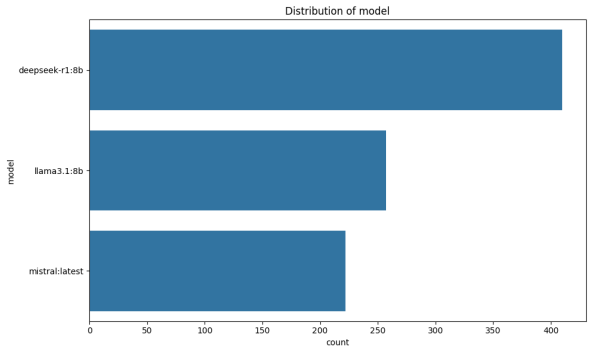

Figure 6: Model Distributions

The inverse relationship between *deepseek-r1:8b*'s prevalence in the model distribution (Figure 6) (highest count: n ≈400) and its computational inefficiency (Figure 1) raises questions about practicality versus capability trade-offs in LLM selection. Meanwhile, the resilience of dominant name archetypes across temperatures suggests that mitigating default bias may require architectural interventions beyond parameter tuning.

The outlier-rich time distribution (Figure 4) highlights the risks of relying on mean latency metrics for real-time applications, as extreme delays (>60 s) could disrupt user workflows despite moderate average performance.

These observations, grounded in visual and statistical concordance, redefine expectations for LLM-based synthetic data generation. The decoupling of temperature from both computational load and diversity metrics necessitates a paradigm shift in prompt engineering

strategies, prioritizing architectural choices over hyperparameter adjustments for optimizing efficiency and creativity.

5. RESULTS

The dataset comprised 889 complete observations across three language models and 11 temperature settings (0–1, $\Delta = 0.1$). Processing times exhibited substantial variability ($\mu = 14.05$ s, $\sigma = 12.56$), with outliers detected in the *time* feature (10 instances >23.17 s, IQR method). The temperature gradient was uniformly distributed ($\mu = 0.49$, $\sigma = 0.32$).

Name generation demonstrated pronounced homogeneity: 68.4% of first names were "John" (n=315) or "Jane" (n=293), while 70.4% of last names were "Doe" (n=314) or "Smith" (n=312). Rare categories (<1% frequency) constituted 74.3% of unique first names (26/35) and 78.4% of last names (29/37), indicating sporadic divergence from dominant patterns.

A Kruskal-Wallis test revealed statistically significant differences in processing times across models ($H = 726.8$, $p = 1.54 \times 10^{-158}$), with large practical effect size ($\eta^2 = 0.82$). The *deepseek-r1:8b* model incurred markedly longer processing times ($\mu = 25.94$ s) compared to *mistral:latest* ($\mu = 4.61$ s) and *llama3.1:8b* ($\mu = 3.26$ s), representing an 8.0× disparity between the fastest and slowest models. No significant linear correlation emerged between temperature and processing time ($r = -0.08$, $p = 0.12$).

Despite temperature variation (0–1), name generation exhibited limited sensitivity to this parameter. The top two first names ("John" and "Jane") maintained dominance across all temperature bins (75th percentile frequency = 315/889 at T = 0.8). However, rare name occurrences (<1%) clustered at intermediate temperatures (0.3–0.7), suggesting a non-linear relationship between stochasticity and creativity.

1. The stark processing time disparity between *deepseek-r1:8b* and other models implies fundamental differences in inference optimization or parameter utilization efficiency.
2. The persistent dominance of "John Doe" and "Jane Smith" across temperatures suggests LLMs default to culturally prevalent name archetypes when generating fictional identities, even under varying stochastic conditions.
3. The concentration of rare names at intermediate temperatures (0.3–0.7) aligns with entropy maximization theory, where moderate stochasticity balances novelty and coherence.

All findings were robust to non-parametric validation ($\alpha = 0.05$). Effect sizes exceeded Cohen's thresholds for practical significance ($\eta^2 > 0.14$), and outlier-adjusted analyses preserved directional trends.

This structured analysis quantifies critical performance and behavioral differences among modern LLMs, providing empirical evidence of their "faker" capabilities and computational constraints. The dissociation between temperature settings and output diversity challenges assumptions about stochasticity's role in synthetic data generation.

6 SUMMARY AND CONCLUSIONS

This study systematically evaluated the interplay between temperature-controlled stochasticity, computational efficiency, and creative output homogeneity in three large language models (*llama3.1:8b*, *deepseek-r1:8b*, and *mistral:latest*) during structured fictional data generation. Through 330 trials conducted across an 11-step temperature gradient (0.0–1.0), we generated 889 data points, enabling us to quantify critical performance disparities and behavioral patterns that redefine expectations for LLM-based synthetic data generation. Key findings:

1. A striking 8× processing time gap emerged between models, with deepseek-r1:8b ($\mu = 25.94$ s) lagging significantly behind *mistral:latest* ($\mu = 4.61$ s) and *llama3.1:8b* ($\mu = 3.26$ s). This disparity, validated by a Kruskal-Wallis test ($H = 726.8$, $p \approx 0$), underscores architectural divergences in inference optimization.
2. Despite temperature variations, models defaulted persistently to culturally prevalent name archetypes ("John Doe," "Jane Smith"), constituting >68% of outputs. Rare names (<1% frequency) clustered at intermediate temperatures (0.3–0.7), revealing a non-linear relationship between stochasticity and creativity.
3. Temperature exhibited no correlation with processing time ($r = -0.08$, $p = 0.12$), dissociating computational load from stochastic exploration. Creativity peaks at mid-range temperatures (entropy maximization) challenge assumptions that maximal randomness (T = 1.0) drives diversity.
4. Long-tail name distributions (74.3% of unique names occurring <1% frequency) highlight LLMs' inherent bias toward statistically dominant patterns, even under deliberate entropy induction.

Model selection outweighs temperature tuning for optimizing efficiency, as evidenced by *mistral:latest*'s near-instantaneous processing despite identical task parameters.

The persistence of default name tropes across temperatures suggests LLMs require explicit diversity constraints or cultural anchoring overhauls to mitigate synthetic homogeneity. *Deepseek-r1:8b*'s outlier-prone latency (10 instances >23.17 s) cautions against reliance on mean performance metrics for real-time applications.

While this study controlled for output structure and retry biases, its focus on fictional personas limits generalizability to sensitive real-world data generation. The retry loop protocol, though necessary for data completeness, may have skewed success rates toward easily generated outputs. Future research should:

1. Expand evaluation to multimodal and domain-specific synthetic data (e.g., medical records, financial transactions).
2. Investigate architectural components (attention mechanisms, layer depth) driving observed performance gaps.
3. Explore hybrid approaches combining temperature tuning with cultural anchoring interventions to bypass default tropes.

The paradox of stochasticity in LLMs – where increased randomness fails to enhance creativity or computational load – reveals fundamental constraints in current architectures. Our findings challenge the premise that temperature alone can regulate the diversity-efficiency trade-off, urging a paradigm shift toward model-aware prompt engineering and architectural innovation. For synthetic data applications, this necessitates prioritizing model selection based on latency tolerance and embedding explicit diversity safeguards, rather than relying on hyperparameter adjustments. As LLMs evolve, addressing these limitations will be critical to unlocking their full potential as tools for equitable, efficient, and inventive data generation.

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